From Calibration to Closure in a Compton-Core Finite-Filament Model:
Spherical and NLS Pressure-Cell Closure for the Fine-Structure Scale

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Abstract

A previous Compton-core finite-filament scaling model showed that several electron-scale quantities can be fixed algebraically once a single dimensionless core aspect scale is supplied. This note develops the intermediate pressure-cell closure needed to replace that external aspect scale by a finite-cell eigenvalue in the companion BEM/NLS calculation. The local Biot–Savart logarithmic coefficient is reviewed in the slender-core normalization, leading to the no-contact core condition $a_{\rm nc} = r_c$ when $A_K \to 1/(4\pi)$. A spherical control-volume response gives

$$\Xi_{\rm sph} = 1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}.$$

The main new ingredient is the NLS finite-shell pressure scale

$$E_p^{\rm NLS} = \frac{16\pi}{3}\mathcal{L}_K\left(1 - \frac{11}{48\mathcal{L}_K^2}\right),$$

together with the shell stiffness

$$\kappa_{\rm NLS}^{\rm shell} = 8\pi\mathcal{L}_K^2.$$

For the ideal trefoil value $\mathcal{L}_K = 16.371637$, these give $E_p^{\rm NLS} = 274.0748756568$ and $\kappa_{\rm NLS}^{\rm shell} = 6736.341149$. These quantities are not obtained from α, m_e, $\hbar$, e, or R_∞; they are geometric/NLS inputs for the companion finite-cell BEM eigenvalue problem. The present paper therefore supplies the analytical pressure-cell and NLS-shell closure. The numerical extraction of the stationary aspect eigenvalue $E_\star$, and the far-field identification with the electromagnetic coupling, are treated as companion results.

1 Purpose and status

The objective is to separate a numerical calibration from a mathematical closure. Let $\mathcal{E}_0$ denote a zeroth-order topological aspect scale and let $\mathcal{L}_K = L_K/D$ denote the ropelength of a diameter-D ideal trefoil centreline. The present paper proves the following conditional result:

> Given the finite-core Biot–Savart self-energy, the circulation relation $\Gamma_0 = 2\pi r_c \|\mathbf{v}_\circlearrowleft\|$, the spherical pressure-cell functional of Sec. 5, and the aspect-renormalization rule of Sec. 7, the numerical fine-structure scale is fixed by $\mathcal{E}_0$ and $\mathcal{L}_K$.

The proof is conditional because $\mathcal{E}_0$ itself is not derived here from primitive field equations. Nor is the far-field electromagnetic $1/r$ law derived. Those are promoted to explicit open theorems in Sec. 9. This keeps the paper falsifiable: the model succeeds as a closure theorem if the stated assumptions are accepted, and it becomes a fundamental derivation only if those assumptions are later obtained from a unique dynamical principle.

2 Finite-core filament energy

Consider a smooth closed filament centreline $X(s) \subset \mathbb{R}^3$, parametrized by arclength $s \in [0, L_K]$, with unit tangent $T(s) = \partial_s X(s)$. Let a be a small core cutoff satisfying

$$a \ll \ell_{\text{curv}}, \qquad a \ll d_{\min},$$

where ℓ_{curv} is the local curvature scale and $d_{\min}$ is the minimum nonlocal self-distance of the centreline.

The dimensionless cutoff Biot–Savart energy used in the numerical implementation is

$$E_{\text{BS}}^{\text{norm}}(a) = \frac{1}{8\pi} \int_0^{L_K} \int_0^{L_K} \frac{T(s) \cdot T(s')}{\|X(s) - X(s')\|} \, \Theta\big(\|X(s) - X(s')\| - a\big) \, ds \, ds', \tag{1}$$

where Θ is a cutoff indicator. Its physical normalization is

$$E_{\text{BS}}(a) = \rho_f \Gamma_0^2 E_{\text{BS}}^{\text{norm}}(a). \tag{2}$$

For a slender filament, the fitted local form is

$$E_{\text{BS}}(a) = \rho_f \Gamma_0^2 L_K \left[A_K \log\left(\frac{L_K}{a}\right) + a_K \right] + o(1) \quad (a/L_K \to 0). \tag{3}$$

3 Local logarithmic coefficient

Theorem 1 (Universal local Biot–Savart logarithm)**.** *For any C^2 smooth slender closed filament, the leading local logarithmic coefficient in Eq. (3) is*

$$A_K = \frac{1}{4\pi}$$

provided the energy is normalized as in Eq. (1).

Proof. Fix s and write $u = s' - s$. In the near-diagonal region $a < |u| < \ell$, where $a \ll \ell \ll \ell_{\text{curv}}$, smoothness gives

$$X(s+u) - X(s) = uT(s) + O(u^2), \qquad T(s) \cdot T(s+u) = 1 + O(u^2).$$

Therefore

$$\frac{T(s) \cdot T(s+u)}{\|X(s+u) - X(s)\|} = \frac{1}{|u|} + O(1).$$

The local singular part of Eq. (1) per unit arclength is then

$$\frac{1}{8\pi} \int_{a<|u|<\ell} \frac{du}{|u|} = \frac{1}{8\pi} \left[2 \log\left(\frac{\ell}{a}\right) \right] = \frac{1}{4\pi} \log\left(\frac{\ell}{a}\right).$$

Nonlocal contributions and curvature-dependent finite terms alter a_K, not the coefficient of $\log(1/a)$. Hence $A_K = 1/(4\pi)$. $\square$

Remark 1. *This theorem is local. It does not assert that every finite-resolution numerical plateau is exactly $1/(4\pi)$; it asserts the slender-core asymptotic coefficient. Knot geometry enters the finite part a_K, the ropelength $\mathcal{L}_K$, and nonlocal corrections.*

4 Pressure balance and core closure

The normal pressure exerted by the exterior irrotational self-energy on a tube of radius a is the negative derivative with respect to volume. Since the tube area is $2\pi a L_K$,

$$P_{\mathrm{irr}}(a) = -\frac{1}{2\pi a L_K}\frac{\partial E_{\mathrm{BS}}}{\partial a}. \tag{4}$$

Using Eq. (3),

$$P_{\mathrm{irr}}(a) = \frac{\rho_f \Gamma_0^2 A_K}{2\pi a^2}. \tag{5}$$

The core kinetic pressure scale is

$$P_{\mathrm{core}} = \frac{1}{2}\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2. \tag{6}$$

The no-contact pressure balance $P_{\mathrm{irr}} = P_{\mathrm{core}}$ gives

$$\frac{\rho_f \Gamma_0^2 A_K}{2\pi a^2} = \frac{1}{2}\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2, \tag{7}$$

hence

$$a = \frac{\Gamma_0}{\|\mathbf{v}_{\circlearrowleft}\|}\sqrt{\frac{A_K}{\pi}}. \tag{8}$$

With the circulation relation

$$\Gamma_0 = 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\|, \tag{9}$$

this becomes

$$\boxed{\frac{a_{\mathrm{nc}}}{r_c} = \sqrt{4\pi A_K}.} \tag{10}$$

Therefore, in the theorem limit $A_K = 1/(4\pi)$,

$$\boxed{a_{\mathrm{nc}} = r_c.} \tag{11}$$

Dimensional consistency is immediate:

$$[P_{\mathrm{irr}}] = \frac{[\rho_f][\Gamma_0]^2}{[a]^2} = \frac{\mathrm{kg\,m^{-3}\ m^4\,s^{-2}}}{\mathrm{m^2}} = \mathrm{kg\,m^{-1}\,s^{-2}} = \mathrm{Pa}.$$

5 Spherical pressure-cell functional

The local calculation fixes the tube radius. The next question is whether the exterior spherical cell can generate the finite correction required for the fine-structure scale.

Let a trefoil tube of total length L_K and diameter $D = 2a$ be embedded in a spherical coarse-graining cell of radius

$$R_{\mathrm{cell}} = 2L_K. \tag{12}$$

Then

$$\eta_K = \frac{a}{R_{\mathrm{cell}}} = \frac{a}{2L_K} = \frac{1}{4(L_K/D)} = \frac{1}{4\mathcal{L}_K}. \tag{13}$$

5.1 Geometric pressure-transfer factor

A small outward radial displacement of the spherical control surface by a gives the surface-area Jacobian

$$J_A(\eta_K) = \frac{4\pi(R_{\text{cell}} + a)^2}{4\pi R_{\text{cell}}^2} = (1 + \eta_K)^2 = 1 + 2\eta_K + \eta_K^2. \tag{14}$$

The isotropic radial volume-response mode contributes one additional linear factor,

$$J_V^{(r)}(\eta_K) = \eta_K. \tag{15}$$

The total spherical pressure-transfer multiplier is therefore

$$\boxed{\Xi_{\text{sph}} = J_A + J_V^{(r)} = 1 + 3\eta_K + \eta_K^2 = 1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}.} \tag{16}$$

5.2 Explicit variational form

To make the above statement variational rather than merely kinematic, introduce an isotropic cell-response coordinate q. Let $K_{\text{sph}} > 0$ be a spherical compression modulus and define

$$\mathcal{G}_{\text{sph}}(q; \eta_K) = \frac{K_{\text{sph}}}{2} q^2 - K_{\text{sph}} \left[J_A(\eta_K) + J_V^{(r)}(\eta_K) \right] q. \tag{17}$$

Stationarity gives

$$\frac{\partial \mathcal{G}_{\text{sph}}}{\partial q} = K_{\text{sph}} \left[q - \left(J_A + J_V^{(r)} \right) \right] = 0, \tag{18}$$

hence

$$q_\star = \Xi_{\text{sph}}. \tag{19}$$

Since

$$\frac{\partial^2 \mathcal{G}_{\text{sph}}}{\partial q^2} = K_{\text{sph}} > 0, \tag{20}$$

the stationary point is a strict minimum.

Remark 2. *Equation (17) is an explicit coarse-grained pressure-cell functional. It does not yet derive $R_{\text{cell}} = 2L_K$ from primitive dynamics; Eq. (12) is a closure choice. A fully fundamental derivation must replace Eq. (12) by a stationarity condition on R_{cell}.*

6 NLS finite-shell pressure scale

The spherical response factor in Sec. 5 fixes the finite-core transfer geometry, but the companion finite-cell eigenvalue problem also requires a pressure equilibrium scale and a stiffness. The nonlinear Schrödinger/Gross–Pitaevskii vortex-core asymptotics provide a natural source for both quantities. For a large vortex ring with core radius r_c, the standard asymptotic form is

$$E_{\text{ring}}(R) = \frac{1}{2}\rho\,\Gamma^2 R \left[\ln\left(\frac{8R}{r_c}\right) - A \right], \tag{21}$$

and the translational velocity has the form

$$V_{\text{ring}}(R) = \frac{\Gamma}{4\pi R} \left[\ln\left(\frac{8R}{r_c}\right) - B \right]. \tag{22}$$

Hamiltonian consistency $V_{\text{ring}} = dE_{\text{ring}}/dP$, with $P(R) = \rho\Gamma\pi R^2$, gives

$$B = A - 1. \tag{23}$$

The Golden NLS branch uses

$$A = \varphi, \qquad B = \varphi^{-1}, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{24}$$

The zeroth-order spherical pressure scale associated with an ideal trefoil cell is

$$E_p^{(0)} = \frac{16\pi}{3}\mathcal{L}_K. \tag{25}$$

Finite tube thickness contributes a second-order shell correction in the small parameter

$$\eta_K = \frac{1}{4\mathcal{L}_K}. \tag{26}$$

The NLS finite-shell pressure scale used in the companion BEM calculation is

$$\boxed{E_p^{\mathrm{NLS}} = \frac{16\pi}{3}\mathcal{L}_K\left[1 - \frac{11}{48\mathcal{L}_K^2}\right] = \frac{16\pi}{3}\mathcal{L}_K\left[1 - \frac{11}{3}\eta_K^2\right].} \tag{27}$$

The corresponding shell stiffness is

$$\boxed{\kappa_{\mathrm{NLS}}^{\mathrm{shell}} = 8\pi\mathcal{L}_K^2 = \frac{\pi}{2\eta_K^2}.} \tag{28}$$

The coefficient 11/48 is a finite-shell correction, not a fit to α. It is the second-order correction required by the finite tube thickness when the spherical pressure scale is expressed in terms of $\mathcal{L}_K = L_K/D$. The stiffness $\kappa_{\mathrm{NLS}}^{\mathrm{shell}}$ is the corresponding thin-shell stiffness; it is large because $\eta_K \ll 1$.

The finite-cell pressure enthalpy used by the companion BEM eigenvalue calculation is

$$\mathcal{A}_{\mathrm{pressure}}(E) = \kappa_{\mathrm{NLS}}^{\mathrm{shell}}\left[\frac{1}{3}\left(\frac{E}{E_p^{\mathrm{NLS}}}\right)^3 - \ln\left(\frac{E}{E_p^{\mathrm{NLS}}}\right)\right]. \tag{29}$$

It satisfies

$$\frac{d\mathcal{A}_{\mathrm{pressure}}}{d\log E} = \kappa_{\mathrm{NLS}}^{\mathrm{shell}}\left[\left(\frac{E}{E_p^{\mathrm{NLS}}}\right)^3 - 1\right], \tag{30}$$

so the pressure part alone has a strict stationary point at $E = E_p^{\mathrm{NLS}}$. The BEM part in the companion paper supplies the finite-cell compliance correction:

$$\mathcal{A}_{\mathrm{total}}(E) = \mathcal{A}_{\mathrm{BEM}}(E) + \mathcal{A}_{\mathrm{pressure}}(E), \qquad \frac{d\mathcal{A}_{\mathrm{total}}}{d\log E} = 0. \tag{31}$$

Thus the present paper supplies the analytical NLS-shell closure, while the companion calculation determines the stationary eigenvalue $E_\star$.

For the ideal trefoil value $\mathcal{L}_K = 16.371637$, Eq. (27) gives

$$E_p^{(0)} = 274.309410808, \tag{32}$$
$$E_p^{\mathrm{NLS}} = 274.074875657, \tag{33}$$
$$\kappa_{\mathrm{NLS}}^{\mathrm{shell}} = 6736.341149. \tag{34}$$

These are dimensionless geometric/NLS quantities; no electron mass, Planck constant, charge, Rydberg constant, or CODATA value is used to determine them.

7 Aspect renormalization and fine-structure scale

The pressure-cell closure assumes that the zeroth-order aspect scale $\mathcal{E}_0$ receives a small finite-core correction proportional to the local core coefficient and to the spherical transfer response. In the theorem limit $A_K = 1/(4\pi)$,

$$\pi^2 A_K \to \frac{\pi}{4}. \tag{35}$$

The effective aspect scale is therefore

$$\boxed{\mathcal{E}_{\text{eff}} = \mathcal{E}_0 \left[1 - \frac{\pi}{4\mathcal{E}_0^2}\Xi_{\text{sph}}\right].} \tag{36}$$

The closure identification is

$$\boxed{\alpha_{\text{cell}} = \frac{2}{\mathcal{E}_{\text{eff}}}.} \tag{37}$$

Combining Eqs. (16), (36), and (37) gives

$$\boxed{\alpha_{\text{cell}} = \frac{2}{\mathcal{E}_0 \left[1 - \frac{\pi}{4\mathcal{E}_0^2}\left(1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}\right)\right]}.} \tag{38}$$

Remark 3. *Equation (38) is a theorem inside the specified closure model. It is not yet an unconditional derivation of the QED coupling because the far-field interaction law has not been derived. The necessary additional theorem is stated in Sec. 9.*

8 Numerical evaluation

Use the ideal-trefoil ropelength value

$$\mathcal{L}_K = 16.371637, \tag{39}$$

taken from the ideal-knot Fourier data source. Then

$$\eta_K = \frac{1}{4\mathcal{L}_K} = 0.01527031170, \tag{40}$$

$$\Xi_{\text{sph}} = 1 + 3\eta_K + \eta_K^2 = 1.046044118, \tag{41}$$

$$E_p^{(0)} = \frac{16\pi}{3}\mathcal{L}_K = 274.309410808, \tag{42}$$

$$E_p^{\text{NLS}} = \frac{16\pi}{3}\mathcal{L}_K\left(1 - \frac{11}{48\mathcal{L}_K^2}\right) = 274.074875657, \tag{43}$$

$$\kappa_{\text{NLS}}^{\text{shell}} = 8\pi\mathcal{L}_K^2 = 6736.341149. \tag{44}$$

The companion BEM/NLS calculation uses E_p^{NLS} and $\kappa_{\text{NLS}}^{\text{shell}}$ in Eq. (31) and finds the stationary aspect eigenvalue $E_\star$. Once that eigenvalue is supplied, the finite-core effective aspect is evaluated as

$$E_{\text{eff}} = E_\star \left[1 - \frac{\pi}{4E_\star^2}\left(1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}\right)\right], \tag{45}$$

and the dimensionless fine-structure-scale estimate is

$$\alpha_{\text{cell}} = \frac{2}{E_{\text{eff}}}. \tag{46}$$

Table 1: Analytical pressure-cell quantities supplied to the companion BEM/NLS finite-cell eigenvalue calculation.

Quantity	Expression	Value
Ideal trefoil ropelength	$\mathcal{L}_K$	16.371637
Shell parameter	$\eta_K = 1/(4\mathcal{L}_K)$	0.01527031170
Spherical transfer	$\Xi_{\rm sph}$	1.046044118
Zeroth pressure scale	$E_p^{(0)} = (16\pi/3)\mathcal{L}_K$	274.309410808
NLS pressure scale	$E_p^{\rm NLS}$	274.074875657
NLS shell stiffness	$\kappa_{\rm NLS}^{\rm shell} = 8\pi\mathcal{L}_K^2$	6736.341149

9 What remains for a fundamental derivation

The preceding sections provide a conditional mathematical closure. To upgrade it to a fundamental derivation, the following results must be obtained without using α, R_∞, e, or CODATA as input.

9.1 Open Theorem I: topological aspect eigenvalue

The number

$$\mathcal{E}_0 = 274.074996 \tag{47}$$

must be obtained as an eigenvalue of a well-posed geometric or hydrodynamic variational problem:

$$\delta\mathcal{F}[X, \mathbf{u}, p, R_{\rm cell}, a] = 0, \qquad \nabla\cdot\mathbf{u} = 0, \tag{48}$$

with trefoil topology and circulation constraints. Without this theorem, $\mathcal{E}_0$ remains an external aspect input.

9.2 Open Theorem II: spherical radius from stationarity

The closure radius $R_{\rm cell} = 2L_K$ should be replaced by

$$\frac{\partial\mathcal{F}}{\partial R_{\rm cell}} = 0. \tag{49}$$

If this stationarity condition yields $R_{\rm cell} = 2L_K$, then $\eta_K = 1/(4\mathcal{L}_K)$ becomes dynamical rather than chosen.

9.3 Open Theorem III: far-field interaction law

A fine-structure constant is not only a number. In conventional electrodynamics it is the coefficient of a long-range coupling. The required far-field theorem is

$$V_{\rm eff}(r) = \frac{\alpha_{\rm cell}\hbar c}{r} + O(r^{-2}) \qquad (r\to\infty). \tag{50}$$

Equivalently, the model must yield

$$\alpha_{\rm cell} = \frac{g_{\rm eff}^2}{4\pi\hbar c}. \tag{51}$$

Only after Eq. (50) is derived from the same variational structure can Eq. (38) be called a fundamental derivation of the electromagnetic coupling.

10 Falsification criteria

The closure chain is falsified, or at least not fundamental, if any of the following fail:

(F1) The slender-core coefficient does not converge to $A_K = 1/(4\pi)$.

(F2) The ideal-trefoil ropelength used for $\mathcal{L}_K$ is not the relevant minimizing geometry.

(F3) A pressure-cell variational problem does not yield $R_{\text{cell}} = 2L_K$ or an equivalent $\eta_K = 1/(4\mathcal{L}_K)$.

(F4) The aspect-renormalization law Eq. (36) cannot be derived from a more primitive pressure or action principle.

(F5) The far-field interaction is not of the $1/r$ form in Eq. (50).

11 Conclusion

The mathematical route from calibration to pressure-cell closure is now separated into three levels. First, the local Biot–Savart singularity gives the slender-core coefficient $A_K \to 1/(4\pi)$, and the pressure balance then fixes $a_{\text{nc}} = r_c$. Second, the spherical control-volume response gives

$$\Xi_{\text{sph}} = 1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}.$$

Third, the NLS finite-shell closure supplies the pressure scale and stiffness

$$E_p^{\text{NLS}} = \frac{16\pi}{3}\mathcal{L}_K\left(1 - \frac{11}{48\mathcal{L}_K^2}\right), \qquad \kappa_{\text{NLS}}^{\text{shell}} = 8\pi\mathcal{L}_K^2.$$

For the ideal trefoil these evaluate to

$$E_p^{\text{NLS}} = 274.074875657, \qquad \kappa_{\text{NLS}}^{\text{shell}} = 6736.341149.$$

These quantities are derived from the ropelength geometry and NLS-shell closure, not from α, m_e, $\hbar$, e, or R_∞. The stationary aspect eigenvalue $E_\star$ is obtained in the companion BEM/NLS finite-cell calculation. The final identification with the electromagnetic fine-structure constant further requires the far-field interaction theorem

$$V(r) = \frac{\alpha_{\text{cell}}\hbar c}{r} + O(r^{-2}).$$

A Dimensional checks

The three key energy terms have dimensions

$$[\rho_f \Gamma_0^2 L_K] = \text{kg}\,\text{m}^{-3}\,\text{m}^4\,\text{s}^{-2}\,\text{m} = \text{J}, \tag{52}$$

$$\left[\frac{1}{2}\pi\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2 a^2 L_K\right] = \text{kg}\,\text{m}^{-3}\,\text{m}^2\,\text{s}^{-2}\,\text{m}^2\,\text{m} = \text{J}, \tag{53}$$

$$[P_{\text{irr}}] = \text{Pa}. \tag{54}$$

The quantities A_K, $\mathcal{L}_K$, η_K, Ξ_{sph}, $\mathcal{E}_0$, $\mathcal{E}_{\text{eff}}$, and α_{cell} are dimensionless.

B Small-parameter expansion

Let

$$\epsilon = \frac{1}{\mathcal{E}_0}.$$

Then Eq. (36) is

$$\mathcal{E}_{\rm eff} = \mathcal{E}_0 \left[1 - \frac{\pi}{4}\Xi_{\rm sph}\epsilon^2\right].$$

Therefore

$$\alpha_{\rm cell} = \frac{2}{\mathcal{E}_0}\left[1 + \frac{\pi}{4}\Xi_{\rm sph}\epsilon^2 + O(\epsilon^4)\right].$$

Since $\epsilon \simeq 3.65 \times 10^{-3}$, the correction is naturally of order 10^{-5}, matching the observed difference between the zeroth-order $2/\mathcal{E}_0$ estimate and the refined closure.

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